Making Proteomics Accessible:
RokaiXplorer for interactive analysis of phospho-proteomic data

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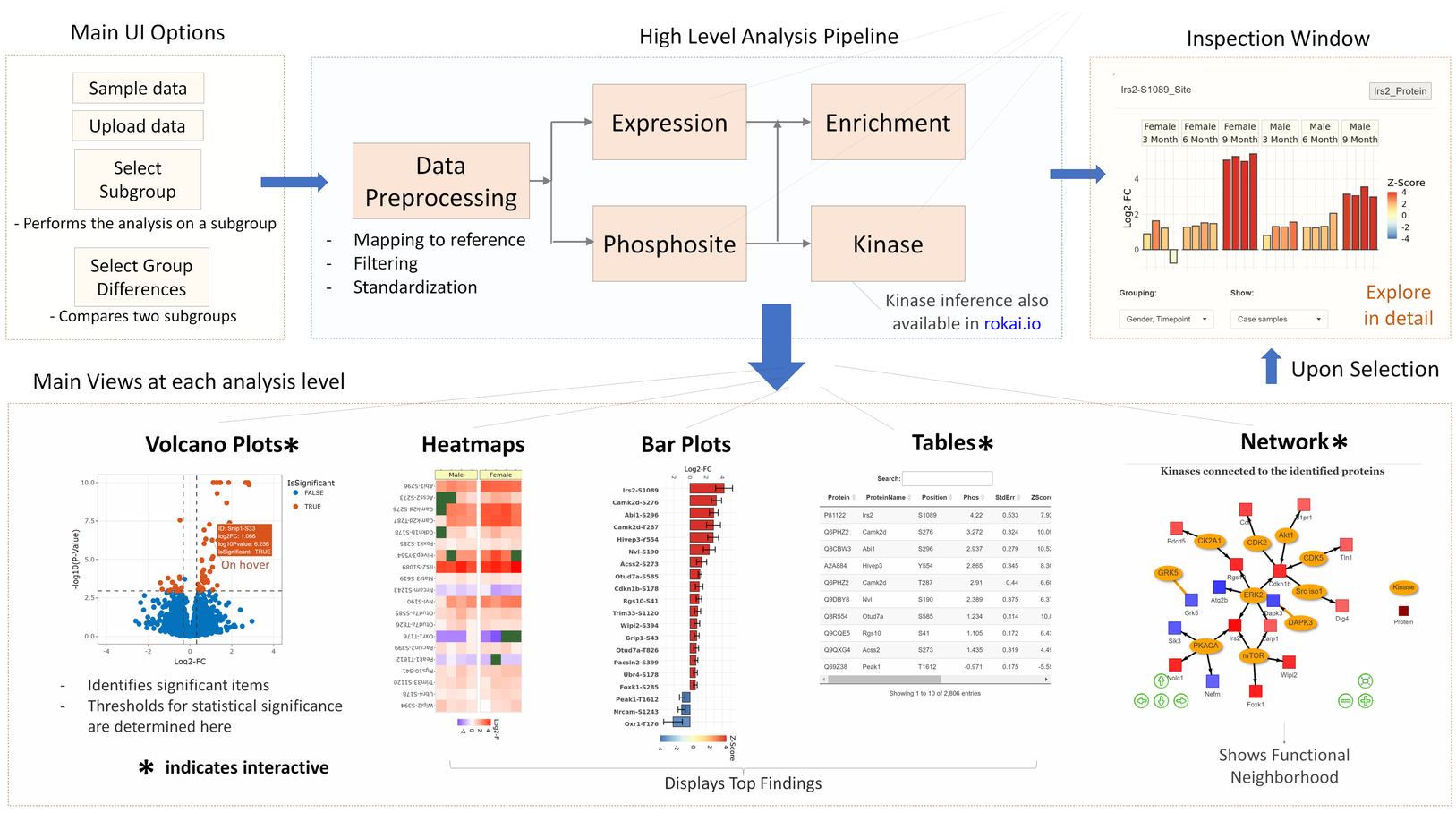
Why RokaiXplorer?

Compared to numerous tools available for other -omics data, there is a lack of user-friendly tools for analyzing and visualizing proteomics and phospho-proteomics data. RokaiXplorer addresses this gap by providing a free, interactive web tool that automates the processing, analysis, and visualization of such data. Specifically tailored for exploratory data analysis, RokaiXplorer empowers researchers to uncover novel biomarkers and potential clinical targets effortlessly!

What can you do with RokaiXplorer?

- **Assess Statistical Significance**: Easily identify significant changes in phosphorylation and protein expression fold changes
- **Enrichment Analysis of GO Terms**: Perform enrichment analysis of gene ontology (GO) terms to gain insights into biological processes and molecular functions
- **Inference of Kinase Activities**: Utilize the RoKAI* algorithm in RokaiXplorer to infer kinase activities and unravel signaling pathway

* Yilmaz et. al. (2021) Robust inference of kinase activity, *Nat. comms.*



Web Application:
http://rokai.io/explorer

Share Your Discoveries!

Beyond individual data analysis, our platform empowers you to seamlessly showcase your findings, insights, and visualizations with the scientific community and beyond.

With the "Share Your Discoveries" feature, deploy your own interactive application in just a few simple steps. Share the link with collaborators, researchers, and stakeholders to explore your data and gain valuable insights. Amplify the impact of your proteomic discoveries and foster collaboration!

AD-Xplorer: An example application on Alzheimer's disease data
https://yilmazs.shinyapps.io/ADXplorer

What do you need to run RokaiXplorer?

Phosphorylation Data

Protein	Position	Sample1	Sample2	Sample3
A1A5B6	S560	NaN	NaN	387918.9
A2A5R2	S1535	684750.2	898782.5	1076999
A2A5R2	S240	250217	388448.4	283210.7
A2A5R2	S355	2110884	1595413	2582802

Uniprot IDs

(Optional) Expression Data

Protein	Sample1	Sample2	Sample3
A2A699	281834.9	305889	305625.8
O88958	NaN	NaN	NaN
Q60930	503886.6	232147.9	305625.8
Q80WJ7	NaN	NaN	NaN

Metadata

RowName	Sample1	Sample2	Sample3
Group	Case	Control	Case
Timepoint	3 Month	3 Month	6 Month
Gender	Male	Female	Female

Specifies the groups of the samples

- Main group that specifies Case/Control status is necessary
- Other groups can be used for filtering in the application

Make sure to try our Interactive Tutorial!

Conflict of interest statement: The authors declare no competing financial interest.